SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0309	**Course Title:**	Data Structures
CO Assessed:	CO2 – Manipulation (BL3, BL4)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	30% of CIA	**Total Marks:**	100
CO2 Statement:	*Implement linear data structures such as stacks, queues, and linked lists, and analyze the efficiency of linear and binary search techniques.*		
Student Name:	R.Rohini sree	**Reg. No.:**	192524471
Section / Batch:	IInd year	**Date:**	17/07/2026

Code of Conduct

I Rachepalli Rohini sree (192524471) *(Name / Reg No)* *certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the Student: Rohinisree

Instructions

- This test contains 80 Analytical problem-solving questions. Total: 100 Marks.
- When a student answer using an Analytical problem-solving, in evaluation the answer should be with following five requirements: Understanding of problem, select appropriate data structure, Design the solution, analyze, Clarity of representation.
- Each student should write 2 questions. No negative marking. Unanswered questions carry zero marks.
- No DTP printouts or electronic devices permitted. They should provide written materials.

Section / Topic	Focus	Q Nos	Marks
Linked Data Structure, Search Data Structure	List Operations, Binary Search	1	50
Stack Data Structure, Queue Data Structures	Stack Notations, Priority Queue	2	50
	GRAND TOTAL:		100 Marks

			Perform the following: ➢ Convert the infix expression into postfix notation. ➢ Show the operator stack after every step. ➢ Evaluate the postfix expression using a stack. ➢ State the final result.
31	192211868	RAMISETTY VENKATA HARSHA VARDHAN	1. Ten documents are requested for printing. Illustrate the mechanism of printer operations. Consider the document number as 1003, 1008, 1010, 1005, 1009, 1004. Identify the suitable data structure and draw the diagram to print the documents. 2. The university announced the selected candidates register number for placement training. The student XXX, Register Number 20142010 wishes to check whether his name is listed or not. The list is not sorted in any order. Identify the searching technique that can be applied and explain the searching steps with suitable procedure. List includes 20142015, 20142033, 20142011, 20142017, 20142010, 20142056, 20142003
32	192524471	RACHEPALLI ROHINISREE	1. Insert the following elements 150, 250, 350, 450, 550 into the double linked list and delete the elements 150 and 250 from the double linked list. Show the linked structure using a node diagram before and after delete operations. 2. A circular queue has a size of 6. Execute the following operations: Enqueue(10), Enqueue(20), Enqueue(30), Dequeue(), Enqueue(40), Enqueue(50), Enqueue(60), Enqueue(70). Draw the queue after each operation and identify the values of the front and rear pointers.
33	192524364	RAGAVI K	1. Convert the expression $((a + b) + c * (d + e) + f) * (g + h)$ to 1) Prefix expression, 2) Postfix expression. 2. Perform the binary search in an array of n elements using a recursion function, in which a function has to call by itself. Use these elements given in sorted order and assume 88 as the key element. [23, 33, 43, 48, 53, 58, 61, 64, 68, 73, 88]
34	192525131	SANKAR R	1. How a queue may be maintained by a circular array with MAXSIZE = 6. Observe that queue always occupies consecutive locations except when it occupies locations at the beginning and at the end of the array. If the queue is viewed as a circular array, this means that it still occupies consecutive locations. The queue is empty when count =0; Perform the following Enqueue and Dequeue operations. a) Initially QUEUE is empty, b) A, B, C are Enqueued, c) A is Dequeued, d) D, E, F are Enqueued, e) B & C are Dequeued, f) G is Enqueued, g) D & E are Dequeued, h) H & I are Enqueued, i) F is dequeued, j) G & H

Rubrics for Calculation

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Understanding	20	Demonstrates complete and precise understanding; identifies all constraints and edge cases	Shows clear understanding with minor gaps	Partial understanding; misses some constraints	Misinterprets problem or lacks clarity
Data Structure Selection	20	Selects optimal data structure with strong justification and comparison	Selects appropriate structure with reasonable justification	Selects somewhat suitable structure with weak reasoning	Incorrect or no data structure selection
Algorithm Design	20	Designs efficient, logically sound, and well-structured algorithm	Algorithm is correct with minor inefficiencies	Algorithm is partially correct or lacks clarity	Algorithm is incorrect or missing
Accuracy of Solution	30	Error-free, well-structured, optimized code/solution	Minor errors but overall functional	Multiple errors; partially working	Non-functional or missing solution
Clarity	10	Exceptionally clear, well-organized, uses diagrams effectively	Clear and organized with minor issues	Somewhat organized but lacks clarity	Poorly organized and unclear

Faculty In-charge	Course Coordinator	Program Director
Signature: ____	Signature: ____	Signature: ____
Date: ____	Date: ____	Date: ____

Question -1

Insert the elements 150, 250, 350, 450, 550 into a doubly Linked List and delete the elements 150 and 250. Show the linked structure before and after deletion.

Introdution

A Doubly Linked List (DLL) is a linear data structure in which each node contains three parts:

- Previous (Prev) pointer - points to the previous node.
- Data - stores the value.
- Next pointer - ~~pointers~~ to the next node.

Unlike a singly linked list, allows traversal in both forward and backward directions, making insertion and deletion operations easier.

Insertion operation:

The following elements are inserted one by one:

150 → 250 → 350 → 450 → 550

Linked Structure After Insertion

NULL ⟷ 150 ⟷ 250 ⟷ 350 ⟷ 450 ⟷ 550 ⟷ NULL

Deletion of Element 150

The element 150 is the first node (head node) of the list. To delete it, the head pointer is moved to the next node, which is 250. The previous Pointer of 250 becomes NULL.

List After Deleting 150

NULL ⟷ 250 ⟷ 350 ⟷ 450 ⟷ 550 ⟷ NULL

Deletion of Element 250

Now 250 becomes the first node. After deleting, it, the head moves to 350, and its previous pointer becomes NULL.

Final Linked Structure.

NULL ⟷ 350 ⟷ 450 ⟷ 550 ⟷ NULL.

Result

After inserting all the elements and deleting 150 and ~~1~~250 the finally doubly linked list is:

NULL ⟷ 350 ⟷ 450 ⟷ 550 ⟷ NULL

Advantages of Doubly Linked List

- It allows traversal in both forward and backward directions.
- Insertion and deletion are easier than is a singly linked List.
- It is widely used in browser history, music playlists, and undo/redo operations.

Question-2

A circular Queue has a size of 6. perform the following operations:

- Enqueue (10)
- Enqueue (20)
- Enqueue (30)
- Dequeue ( )
- Enqueue (40)
- Enqueue (50)
- Enqueue(60).
- Enqueue (70).

Introduction

A circular Queue is a special-type of queue in which the last position is connected to the first position. when the rear reaches the last index, it wraps around to the beginning if there is free space.

The queue follows the FIFO (First In, First out) principle. This means the first element inserted is the first element removed.

Queue size = 6

Initial Queue

[-][-][-][-][-]

Front = -1

Rear = -1

Step-1: Enqueue(10).

Insert 10 into the empty queue.

[10][-][-][-][-][-]

Front=0

Rear=0

Step-2: Enqueue(20).

Insert 20 at the rear.

[10][20][-][-][-][-]

Front = 0

Rear=1

Step-3: Enqueue(30)

Insert 30

[10][20][30][-][-][-]

Front = 0

Rear = 2

[70][20][30][40][50][60]

Front = 1

Rear = 0

Advantages of circular Queue

- It efficiently utilizes memory by reusing empty spaces.
- It avoids the shifiting of elements after deletion.
- It is commonly used in a natural, student-friendly style and network packet management.

Step 4: Dequeue()

Delete the first element (10) Front moves to the next position

[-] [20] [30] [-] [-] [-]

Front = 1

Rear = 2

Step 5: Enqueue (40)

[-] [20] [30] [40] [-] [-]

Front = 1

Rear = 3

Step 6: Enqueue (50).

[-] [20] [30] [40] [50] [-]

Front = 1

Rear = 4

Step 7: Enqueue (60).

[-] [20] [30] [40] [50] [60]

Front = 1

Rear = 5

Step 8: Enqueue (70).

Since it is a circular Queue, the rear wraps around to index 0.